

language on the WWW. Why, and for how long, need this imbalance exist? And remember, with all the attention to multimedia, one is apt to forget that text is still the mainstay of communication! The world needs automated translation—and soon.

Artificial Imbeciles

The MT problem is, of course, within the domain of AI (Artificial Intelligence), amongst other disciplines. MT is very closely related to Natural Language Understanding (NLU)—where a computer attempts to understand what a spoken or written sentence means. Now “understanding” is something that’s been subjected to much philosophical, scientific and pseudo-scientific debate, so we won’t debate it here. But then, stated one way, the goal of AI—as far as the input side is concerned—is indeed to understand. In that light, we can say that MT will advance hand in hand with AI.

But when? MT hasn’t really seen any major breakthroughs since its beginnings in the 50s. In AI, promises have all too often been made and not delivered upon; MT is no exception. Some people say, “It’s all there: the hardware (fast machines), the software (neural networks and fuzzy logic), and it’s just a matter of time before we get good automated translations...” Problem is, it’s turning out to be a really long time.

Some people have noted, though, that other industries that do AI research—robotics, for example—could come up with insights and methods that would further the field. Then, there’s the possibility that since one of the biggest problems in AI is infusing common sense into systems, MT would benefit as bodies of common sense become bigger and better and can be easily “plugged in” to existing systems.

Common sense apart, there’s the problem of knowledge, as noted before. Unlike systems such as chat-bots that require a little knowledge and a lot of common sense, and unlike systems such as chess-playing programs that work on lots of knowledge and heuristics and not too much common sense, MT requires *both* lots of knowledge and lots of common sense!

Why knowledge? We mentioned earlier that a system needs to know that an elephant’s trunk is an unlikely place to shove a shirt into. Such things straddle the border between common sense and knowledge. But there’s pure knowledge to be considered, too: certain words do *not* occur in certain contexts, such as “glycogen” in

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the context of reportage (something on this later). “You” can be “*tum*,” “*aap*,” “*aap log*,” or “*tum log*” in Hindi, depending on context. Cultural and regional differences—both in terms of world reality and word usage—exist between regions and languages respectively.

The problem we’re pointing at is to decide how much knowledge needs to be put in the system, and how to encode it all so the machine translator can put it to good use. And that is going beyond even AI. It involves a collaboration between cognitive scientists, linguists, language and cultural experts, computer scientists, and more.

Methods To The Madness

But talking about understanding, we’re led to one obvious way to try and translate documents, comprising the following steps:

- 1 Understand the source document
- 1 Put it in an intermediate format
- 1 Generate the output in the target language

This is a step ahead from the “direct” method of translation, which just takes a language pair, inputs dictionaries of both, and tries to generate translations without paying attention to context and other nuances. The intermediate-representation approach also has the advantage that once the intermediate “understanding” is generated, you could translate that into any language, not just one.

A third approach is called the “transfer approach,” consisting again of three steps. Here, the first stage converts the source language to an abstract representation; the second maps these representations to *representations* in the target language; and the third stage generates the translation. Note that the system doesn’t exactly try and “understand” what’s going on in the source language; there are abstract representations involved for both languages, which incorporate such things as context and real-world knowledge.

But what seems really promising now is the statistical approach, which we’ll soon talk about.

Google Everywhere!

In August 2005, in a US government-run test, Google’s translation application beat technology

